

The diagram is a plot of specific enthalpy h (in $\frac{\text{kJ}}{\text{kg}}$) versus specific entropy s (in $\frac{\text{kJ}}{\text{kgK}}$). The x-axis ranges from 300 to 700 $\frac{\text{kJ}}{\text{kg}}$, and the y-axis ranges from 0 to 1000 $\frac{\text{kJ}}{\text{kg}}$. The plot shows the saturation dome for water, with the critical point at approximately $h = 2039 \frac{\text{kJ}}{\text{kg}}$ and $s = 4.01 \frac{\text{kJ}}{\text{kgK}}$. The liquid region is to the left of the dome, and the vapor region is to the right. The process 2-2'-3-4-5-6-7-8-9-10-11-12-13-14-15-16-17-18-19-20-21-22-23-24-25-26-27-28-29-30-31-32-33-34-35-36-37-38-39-40-41-42-43-44-45-46-47-48-49-50-51-52-53-54-55-56-57-58-59-60-61-62-63-64-65-66-67-68-69-70-71-72-73-74-75-76-77-78-79-80-81-82-83-84-85-86-87-88-89-90-91-92-93-94-95-96-97-98-99-100 is shown as a red line, starting at point 2 (saturated liquid at 10 bar), going to point 2' (superheated vapor at 10 bar), then to point 3 (saturated vapor at 1 bar), point 4 (superheated vapor at 1 bar), point 5 (saturated liquid at 1 bar), point 6 (subcooled liquid at 1 bar), point 7 (saturated liquid at 10 bar), point 8 (superheated vapor at 10 bar), point 9 (saturated vapor at 1 bar), point 10 (superheated vapor at 1 bar), point 11 (saturated liquid at 1 bar), point 12 (subcooled liquid at 1 bar), point 13 (saturated liquid at 10 bar), point 14 (superheated vapor at 10 bar), point 15 (saturated vapor at 1 bar), point 16 (superheated vapor at 1 bar), point 17 (saturated liquid at 1 bar), point 18 (subcooled liquid at 1 bar), point 19 (saturated liquid at 10 bar), point 20 (superheated vapor at 10 bar), point 21 (saturated vapor at 1 bar), point 22 (superheated vapor at 1 bar), point 23 (saturated liquid at 1 bar), point 24 (subcooled liquid at 1 bar), point 25 (saturated liquid at 10 bar), point 26 (superheated vapor at 10 bar), point 27 (saturated vapor at 1 bar), point 28 (superheated vapor at 1 bar), point 29 (saturated liquid at 1 bar), point 30 (subcooled liquid at 1 bar), point 31 (saturated liquid at 10 bar), point 32 (superheated vapor at 10 bar), point 33 (saturated vapor at 1 bar), point 34 (superheated vapor at 1 bar), point 35 (saturated liquid at 1 bar), point 36 (subcooled liquid at 1 bar), point 37 (saturated liquid at 10 bar), point 38 (superheated vapor at 10 bar), point 39 (saturated vapor at 1 bar), point 40 (superheated vapor at 1 bar), point 41 (saturated liquid at 1 bar), point 42 (subcooled liquid at 1 bar), point 43 (saturated liquid at 10 bar), point 44 (superheated vapor at 10 bar), point 45 (saturated vapor at 1 bar), point 46 (superheated vapor at 1 bar), point 47 (saturated liquid at 1 bar), point 48 (subcooled liquid at 1 bar), point 49 (saturated liquid at 10 bar), point 50 (superheated vapor at 10 bar), point 51 (saturated vapor at 1 bar), point 52 (superheated vapor at 1 bar), point 53 (saturated liquid at 1 bar), point 54 (subcooled liquid at 1 bar), point 55 (saturated liquid at 10 bar), point 56 (superheated vapor at 10 bar), point 57 (saturated vapor at 1 bar), point 58 (superheated vapor at 1 bar), point 59 (saturated liquid at 1 bar), point 60 (subcooled liquid at 1 bar), point 61 (saturated liquid at 10 bar), point 62 (superheated vapor at 10 bar), point 63 (saturated vapor at 1 bar), point 64 (superheated vapor at 1 bar), point 65 (saturated liquid at 1 bar), point 66 (subcooled liquid at 1 bar), point 67 (saturated liquid at 10 bar), point 68 (superheated vapor at 10 bar), point 69 (saturated vapor at 1 bar), point 70 (superheated vapor at 1 bar), point 71 (saturated liquid at 1 bar), point 72 (subcooled liquid at 1 bar), point 73 (saturated liquid at 10 bar), point 74 (superheated vapor at 10 bar), point 75 (saturated vapor at 1 bar), point 76 (superheated vapor at 1 bar), point 77 (saturated liquid at 1 bar), point 78 (subcooled liquid at 1 bar), point 79 (saturated liquid at 10 bar), point 80 (superheated vapor at 10 bar), point 81 (saturated vapor at 1 bar), point 82 (superheated vapor at 1 bar), point 83 (saturated liquid at 1 bar), point 84 (subcooled liquid at 1 bar), point 85 (saturated liquid at 10 bar), point 86 (superheated vapor at 10 bar), point 87 (saturated vapor at 1 bar), point 88 (superheated vapor at 1 bar), point 89 (saturated liquid at 1 bar), point 90 (subcooled liquid at 1 bar), point 91 (saturated liquid at 10 bar), point 92 (superheated vapor at 10 bar), point 93 (saturated vapor at 1 bar), point 94 (superheated vapor at 1 bar), point 95 (saturated liquid at 1 bar), point 96 (subcooled liquid at 1 bar), point 97 (saturated liquid at 10 bar), point 98 (superheated vapor at 10 bar), point 99 (saturated vapor at 1 bar), point 100 (superheated vapor at 1 bar).

The diagram is a pressure-enthalpy (p-h) plot for R134a. The vertical axis represents pressure (p) in bar on a logarithmic scale from 10^0 to 10^2 . The horizontal axis represents enthalpy (h) in $\frac{kJ}{kg}$ from 450 to 700. The plot includes saturation curves (dashed lines) and various thermodynamic property lines: isotherms (solid lines labeled with temperatures like 30.0 °C, 60.0 °C, 120.0 °C, 150.0 °C), isochors (dotted lines labeled with specific volumes like 0.003 $\frac{m^3}{kg}$, 0.01 $\frac{m^3}{kg}$, 0.1 $\frac{m^3}{kg}$, 0.3 $\frac{m^3}{kg}$), and isentropes (solid lines labeled with entropy values like 0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0). A red cycle is highlighted with four points: 31 (superheated vapor at ~620 $\frac{kJ}{kg}$, ~5 bar), 32 (superheated vapor at ~680 $\frac{kJ}{kg}$, ~20 bar), 33 (subcooled liquid at ~480 $\frac{kJ}{kg}$, ~20 bar), and 34 (subcooled liquid at ~480 $\frac{kJ}{kg}$, ~5 bar). The cycle consists of four processes: 31-32 (compression), 32-33 (condensation), 33-34 (expansion), and 34-31 (evaporation).